

Jonatan Langlet, PhD

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Swedish

Research & Engineering - Advanced Systems

Work Experience

- Sep 2024– **Postdoctoral Researcher**, KTH ROYAL INSTITUTE OF TECHNOLOGY
- Researching large-scale data analysis systems and algorithm design
 - Supervising Master's projects for system-on-a-chip development
- June-July 2023 **R&D Engineering Internship**, INTERNET INITIATIVE JAPAN
- On-site prototype development of an intelligent network fabric for a Japanese ISP
- Mar-Oct 2022 **Research Fellow**, HARVARD UNIVERSITY
- Led research into hyperscale network monitoring techniques for fine-grained telemetry
 - I still regularly collaborate with Prof. Michael Mitzenmacher and Prof. Minlan Yu at Harvard Systems+Theory
- 2020–2022 **Lab- and Teaching Assistant**, QUEEN MARY UNIVERSITY OF LONDON
- Teaching Distributed Systems for Master's students
- 2019–2020 **Research Assistant**, KARLSTAD UNIVERSITY
- Researching in-network acceleration for machine learning and 5G core networks
- 2019 **Teaching Assistant**, KARLSTAD UNIVERSITY
- Teaching Data Structures & Algorithms for Bachelor's students
- 2011–2016 **Owner/Developer**, TANAPPS
- Sole proprietorship focusing on web application development and software development
 - Installation and maintenance of a virtualization cluster for hosting clients' applications
- 2011–2013 **IT Technician**, FOUREDGE
- Server installation/maintenance with a main focus on Linux and virtualization
 - Sysadmin consultant for clients, designing and maintaining IT environments
 - Second line helpdesk support, troubleshooting server issues

Education

- 2020–2024 **PhD, Networking Systems**, QUEEN MARY UNIVERSITY OF LONDON
- Researched technologies for high-speed streaming data collection and processing
 - Focus on algorithms, data structures, and systems at the intersection of software and hardware
- 2019–2020 **MSc, Computer Science**, KARLSTAD UNIVERSITY
- Graduated one year early due to receiving PhD offers
 - Grade: Highest average of my cohort
 - Thesis: Offloading Virtual Network Functions–Hierarchical Approach
- 2016–2019 **BSc, Computer Engineering**, KARLSTAD UNIVERSITY
- Grade: Highest average of my cohort
 - Thesis: Towards Machine Learning Inference in the Data Plane

Publications

- 2024 **ACM SIGMETRICS**, *Lightweight Acquisition and Ranging of Flows in the Data Plane*
Andrea Monterubbiano, **Jonatan Langlet**, Stefan Walzer, Gianni Antichi, Pedro Reviriego Vasallo, and Salvatore Pontarelli

- 2023 **ACM SIGCOMM**, *Direct Telemetry Access*
Jonatan Langlet, Ran Ben Basat, Gabriele Oliaro, Michael Mitzenmacher, Minlan Yu, and Gianni Antichi
- 2022 **IEEE TMC**, *Hybrid P4 Programmable Pipelines for 5g gNodeB and User Plane Functions*
Suneet Kumar Singh, Christian Esteve Rothenberg, **Jonatan Langlet**, Andreas Kassler, Péter Vörös, Sándor Laki, and Gergely Pongrácz
- 2021 **ACM HotNets**, *Zero-CPU Collection with Direct Telemetry Access*
Jonatan Langlet, Ran Ben Basat, Sivaramakrishnan Ramanathan, Gabriele Oliaro, Michael Mitzenmacher, Minlan Yu, and Gianni Antichi
- 2019 **IEEE EuroP4**, *Towards Neural Network Inference on Programmable Switches*
Jonatan Langlet, Andreas Kassler, and Deval Bhamare

Academic Community Work

- 2025 **TPC Member**, *TMA Conference*
- 2024 **External Reviewer**, *ACM CoNEXT*
- 2023– **Invited Paper Reviewer**, *IEEE/ACM Transactions on Networking*
- 2023– **Invited Paped Reviewer**, *IEEE Transactions on Network and Service Management*

Awards, Certificates, and Extra Courses

- 2025 **Best Poster Award**, Awarded at the Digital Futures Open Research Day
- 2023 **Digital Futures Postdoctoral Fellowship**, Two years of independent research funding
- 2022 **Brendan Murphy Prize winner**, Awarded by the UK networking community
- 2022 **SIGCOMM Travel Grant**, Including accommodation, conference registration, and travel
- 2021 **NVIDIA Hardware Grant**, I was contacted regarding a BlueField-2 DPU donation
- 2021 **NVIDIA DPU Course & Hackathon**, A crash course on a new programmable NIC
- 2020 **Xilinx Adaptive Compute PhD school**, A week-long course on FPGA development
- 2019 **Barefoot Academy BA-102**, A week-long course on P4-programmable Tofino switches
- 2012 **ITIL Foundation Certification**
- 2011 **Cisco CCNA Discovery**

Skills

Areas	Algorithm Development, System on a Chip, Network Monitoring, Data Planes, Machine Learning, Time-Series Analysis, Switching Architectures, Embedded Systems
Programming	Python, P4, C, C++, Micro-C, Java, PHP, C#, Bash, among others
Software	Linux, Git, DPDK, VMware ESXi, MySQL, AWS, among many others
Soft Skills	Collaborative, Fast Learner, Organized, Goal Oriented, Methodical

Languages

Swedish	Mothertongue
English	Fluent

Main working language